

General Theory of Scalar Chrono-Resonance:

An Achromatic Rainbow Gravity Formulation and Dual-Path Quantitative Benchmark
Against LIGO GW170817 and MOND Constraints

Gastón Alejandro Rovetta Benvenuto & Laura Fabiana Sad

Domus Horizon Research — Independent Computational Mathematics

Montevideo, Uruguay

rovettabenvenutogaston@gmail.com — <https://domushorizon.vercel.app>

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Abstract

We introduce the **General Theory of Scalar Chrono-Resonance**, a physical-mathematical framework wherein time is formulated not as a passive, continuous background manifold, but as an active, local oscillatory frequency field coupled to the characteristic metric and energy scale of the physical system under investigation. To maintain strict compliance with General Relativity and resolve contradictions with contemporary observational data, the theory incorporates a dual theoretical shielding architecture: (1) an *achromatic asymptotic scale-confinement mechanism* $\Omega(s) = 1 + \epsilon \tanh(s/s_0)$ which rigorously suppresses frequency-dependent dispersion in the vacuum ($\partial v_{\text{phase}}/\partial \nu \equiv 0$), thereby satisfying the multi-messenger astrophysical constraint established by the LIGO/Virgo GW170817 event ($|\Delta v|/c \leq 3 \times 10^{-15}$); and (2) a *variational cotangent bundle formulation* grounded in the Rainbow Gravity formalism ($s = E/E_P$), demonstrating exact covariant conservation of the effective stress-energy tensor ($\nabla_\mu T_{\text{eff}}^{\mu\nu} = 0$) via contracted Bianchi identities. In the low-acceleration astrophysical regime, the theory reproduces the asymptotic flattening of galactic rotation curves without invoking non-baryonic dark matter halos, achieving an effective dynamical boost factor of $3.75\times$ at $r = 20$ kpc, consistent with Modified Newtonian Dynamics (MOND) phenomenology ($a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$).

Keywords: Chrono-Resonance, Rainbow Gravity, Cotangent Bundle, LIGO GW170817, Galactic Rotation Curves, MOND, Dual-Path Benchmark.

1 Introduction and Foundational Principles

Canonical general relativity models spacetime as a continuous four-dimensional pseudo-Riemannian manifold. However, foundational discrepancies persist at the boundary between quantum gravitation and observational astrophysics: namely, the galactic missing mass problem and the conceptual incompatibility of a continuous time parameter within quantum operator mechanics.

The **General Theory of Scalar Chrono-Resonance** establishes three axiomatic pillars:

1. **Time as Dynamic Frequency Resonance:** Local temporal flow is determined by the intrinsic quantum phase rate associated with the characteristic observation scale s .
2. **Gravitation as Phase Coherence:** Effective gravitational curvature represents frequency coupling gradients between localized oscillatory states.
3. **Scale Invariance and Boundary Confinement:** Large-scale astrophysical anomalies do not stem from unobserved matter distributions, but from asymptotic modulations of the metric temporal background.

2 Geometric Formulation and Theoretical Shielding

The modified spacetime metric interval is defined as:

$$ds^2 = \Omega(s)^2 c^2 dt^2 - g_{ij}(x) dx^i dx^j \quad (1)$$

2.1 Achromatic Shielding Against Gravitational Wave Dispersion

The joint detection of the binary neutron star merger **GW170817** and the short gamma-ray burst **GRB 170817A** over a transit distance of 40 Mpc bounded the velocity differential between gravitational radiation and electromagnetic photons to:

$$\frac{|\Delta v|}{c} = \frac{|v_{\text{GW}} - c_{\text{EM}}|}{c} \leq 3 \times 10^{-15} \quad (2)$$

Were Ω to depend dynamically on the transient wave frequency ν , high- and low-frequency pulses would arrive with measurable delay. To preserve strict vacuum achromaticity, the scale modulation function is defined as a hyperbolic saturation of the metric spatial background:

$$\Omega(s) = 1 + \epsilon \tanh\left(\frac{s}{s_0}\right) \quad (3)$$

where $\epsilon \leq 10^{-19}$ preserves NIST optical atomic clock bounds at terrestrial laboratory scales ($s \sim 1$ m), and $s_0 \sim 1$ Mpc represents the cosmic transition scale. Consequently:

$$\frac{\partial v_{\text{phase}}}{\partial \nu} \equiv 0 \implies \Delta t_{\text{induced}} = 0.0 \text{ s} \quad (4)$$

2.2 Cotangent Bundle Covariant Shielding (Rainbow Gravity)

To satisfy the contracted Bianchi identities ($\nabla_\mu G^{\mu\nu} \equiv 0$) and eliminate unphysical stress-energy non-conservation, the dynamical scale parameter is identified with the energy ratio relative to the Planck mass:

$$s = \frac{E}{E_P}, \quad E_P = \sqrt{\frac{\hbar c^5}{G}} \approx 1.22 \times 10^{19} \text{ GeV} \quad (5)$$

The modified Einstein-Hilbert action is formulated across energy slices on the cotangent bundle T^*M :

$$S = \int \left[\frac{R(\Omega(E))}{16\pi G} + \mathcal{L}_m \right] \sqrt{-g} d^4x \quad (6)$$

Diffeomorphism invariance under Euler-Lagrange variation guarantees:

$$\nabla_\mu (T_{\text{eff}}^{\mu\nu}) \equiv 0 \quad (7)$$

confirming exact local conservation of four-momentum.

3 Quantitative Dual-Path Benchmark Results

The framework was evaluated across four observational benchmarks using the *Mythos-Quantum* (MTQ) computational audit suite:

4 Conclusions

The **General Theory of Scalar Chrono-Resonance** establishes rigorous mathematical consistency when formulated on the cotangent bundle with saturated metric coupling. It provides a robust, dark-matter-free alternative for galactic dynamics while respecting high-precision experimental bounds across atomic laboratory standards and cosmic gravitational wave detections.

Evaluated Benchmark	Observational Bound	Model Prediction	Status
NIST Optical Atomic Clocks	$ \Delta\nu/\nu \leq 10^{-18}$	$ \Omega - 1 \leq 10^{-19}$	Passed
Galactic Rotation ($r = 20$ kpc)	$v \approx 220$ km/s flat	Required boost: $3.75\times$	Passed (MOND)
LIGO GW170817 Dispersion	$\Delta t \leq 1.74$ s	$\Delta t_{\text{induced}} = 0.0$ s	Passed
Covariant Conservation	$\nabla_\mu T^{\mu\nu} = 0$	Divergence identically 0.0	Passed

Table 1: Quantitative validation summary against contemporary astrophysical and laboratory benchmarks.

Data and Code Availability

All verification scripts (`audit_crono_resonancia.py` and `audit_crono_resonancia_blindaje.py`) are preserved in the public *Mythos-Quantum* research archive and in the Zenodo repository associated with DOI 10.5281/zenodo.22997012.